MIL-G-48246 (PA)
26 November 1973
SUPERSESSION
(see 6.5)

MILITARY SPECIFICATION

GLYCERYL ESTER OF 12-HYDROXY STEARIC ACID

1. SCOPE

1.1 Scope. This specification covers glyceryl ester of 12-hydroxy stearic acid for use in ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-D-705 -	Drum, Metal Shipping, Steel (over 12 and under 55 Gallons)
PPP-D-711 -	Drum, Metal Shipping, Steel, Lightweight (55 Gallon)
PPP-D-723 -	Drum, Fiber
PPP-D-729 -	Drum, Metal, 55 Gallon (For Shipment of Non-Corrosive Material)

STANDARDS

MILITARY

MIL-STD-105 -	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 -	Inspection Terms and Definitions
MIL-STD-129 -	Marking for Shipment and Storage
MIL-STD-650 -	Explosive: Sampling, Inspection and Testing
MIL-STD-1168-	Lot Numbering of Ammunition

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC: 6810

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D127 - Drop Melting Point of Petroleum Wax, Including Petrolatum, Test for,
D 1321 - Needle Penetration of Petroleum Waxes, Test for,
D 1957 - Hydroxyl Value of Fatty Oils and Acids, Test for,
D 1959 - Iodine Value of Drying Oils and Fatty Acids, Test for
D 1962 - Saponification Value of Drying Oils, Fatty Acids and Polymerized Fatty Acids, Test for
D 1980 - Acid Value of Fatty Acids and Polymerized Fatty Acids, Test for,
D 2669 - Apparent Viscosity of Petroleum Waxes Compounded with Additives (Hot Melts)
E 300 - Industrial Chemicals, Recommended Practice for Sampling

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19130)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies).

3. REQUIREMENTS

3.1 Chemical requirements. Glyceryl ester of 12-hydroxy stearic acid shall comply with the chemical requirements specified in Table I when tested in accordance with the specified test method listed in Section 4.3.3, Table II.

TABLE I
CHEMICAL REQUIREMENTS

Acid Number, max	4.0
Alkalinity, % max	None

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Saponification Number (range)	175-185
Hydroxyl Value, min	154
Melting Point range °C	83-88
Iodine Number, max.	4
Penetration(0.1mm),@77°F, min	2
Viscosity, Centipoise at 200°F max	24
at 250°F min	16
Exudation, % max (160°F)	3.0

3.2 Reactivity at 120°C. The reactivity value shall not exceed 3.0 ml of gas over and above that generated by the controls when tested as specified in 4.4.10 (see 6.4).

3.3 First article testing. This specification makes provision for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein.

4.1.1 Submission of product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attest that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Specification number and date, together with identification and date of changes thereto.

c. Certificates of analysis on all materials used directly by Government specifications shall be made available

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upon request by the Contracting Officer.

d. A statement that there has been no change to process or raw materials or any other change which could have altered the characteristics of the product from the product which was originally submitted for qualification

e. Quantity of product in the lot.

f. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission. Prior to initiation of regular production the contractor shall submit a first article sample consisting of (50) fifty pounds of glyceryl ester of 12-hydroxy stearic acid in accordance with instructions issued by the Contracting Officer for evaluation in accordance with paragraph 4.2.2. All samples submitted shall have been produced by the contractor using the same production process, procedures, and equipment as will be used in fulfilling the contract. All materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis. A first article quantity, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing specification or source of supply as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.3 and 4.4 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection. If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspections upon any failure of a sample to

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comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation. A lot shall consist of that quantity of glyceryl ester of 12-hydroxy stearic acid produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of material that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordance with MIL-STD-105 (or MIL-STD-1235 when applicable). The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.2 Examination. Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect.

4.3.2.1 Container sealed

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Container damaged to the extent that the contents are exposed, or likely to become exposed in subsequent handling	Visual	01001
Minor:	AQL 0.65 percent		
201.	Marking misleading or unidentifiable	Visual	01002

4.3.3 Sampling. A representative sample of approximately 5 pounds of glyceryl ester of 12-hydroxy stearic acid shall be selected from each lot of material submitted for acceptance. Use the method described in ASTM E300 for solids for withdrawing the samples. The sample shall be tested for the following properties in accordance with the test methods of Section 4.4. If the sample fails to meet any of the

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specified requirements, the lot shall be rejected (see Table II). Unless otherwise specified in the contract or purchase order, the test for reactivity shall be performed on the first article sample only.

TABLE II

Test	Requirements	Test Method	Defect Class	Code No.
Acid Number	3.1	4.4.1	Major	02001
Alkalinity	3.1	4.4.2	Major	02002
Saponification	3.1	4.4.3	Major	02003
Hydroxyl value	3.1	4.4.4	Major	02004
Melting point	3.1	4.4.5	Major	02005
Iodine number	3.1	4.4.6	Major	02006
Penetration	3.1	4.4.7	Major	02007
Viscosity	3.1	4.4.8	Major	02008
Exudation	3.1	4.4.9	Major	02009
Reactivity at 120°C	3.2	4.4.10	Major	02010

4.4 Test methods and procedures

4.4.1 Determination of acid number. The acid number shall be determined in accordance with ASTM D1980.

4.4.2 Alkalinity. The alkalinity shall be determined in accordance with method 102.3 of MIL-STD-650 using benzene as the solvent.

4.4.3 Determination of saponification number. The saponification number shall be determined in accordance with ASTM D1962.

4.4.4 Determination of hydroxy value. The hydroxyl value shall be determined in accordance with ASTM D1957.

4.4.5 Determination of melting point. The melting point shall be determined in accordance with ASTM D127.

4.4.6 Determination of iodine number. The iodine number shall be determined in accordance with ASTM D 1959, except that chloroform shall be used instead of carbon tetrachloride.

4.4.7 Penetration. The penetration shall be determined in accordance with ASTM D 1321.

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4.4.8 Viscosity. The viscosity shall be determined in accordance with ASTM D2669.

4.4.9 Exudation. The sample shall be melted at approximately 220°F. When completely molten, the wax shall be cooled with stirring to about 10°F above its drop melting point and cast into a glass mold, 1 inch inside diameter and 0.5 inch high. The glass mold shall rest on a smooth 0.5 inch thick aluminum plate and at the time of casting, both plate and mold shall be at 70° ± 10°F. Under these conditions the sample will freeze in 3 to 5 minutes. The sample shall be removed from the mold and placed on 20 layers of filter paper, cut in 1 1/2 inch diameter circles. An 80 gm brass weight, 1 inch diameter, shall be placed on top of the wax casting. This assembled sample shall then be placed on a wire screen and stored in a forced draft oven for 24 hours at 160°F ± 1/2°F. Before assembly, the brass weight shall be weighed, and then reweighed with the wax casting. After hot storage, wax and brass weight shall be removed from the filter paper while still warm, no attempt being made to separate brass weight and wax and allowed to cool upside down (the wax resting on the brass weight). When cool, the wax and brass weight shall be weighed together. The difference in weights shall be calculated as percent lost during storage. The test shall be run in triplicate and close agreement of the individual results shall be obtained.

$$\% \text{ Exudation} = \frac{A-C}{A-B} \times 100$$

A = weight of wax plus brass weight before exudation test
B = weight of the brass weight
C = weight of wax plus brass weight after test exudation

4.4.10 Determination of Reactivity at 120°C. The reactivity of the glyceryl ester of 12-hydroxy stearic acid shall be determined with both RDX and TNT in accordance with Method 504.1 of MIL STD 650. The reactivity value is defined as the volume of gas produced by the wax explosive mixture minus the sum of the volume of gases individually produced by the wax and the explosive. (see 6.4)

5. PREPARATION FOR DELIVERY

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5.1 Preservation and packaging not applicable.

5.2 Packing

5.2.1 Level A. The glyceryl ester of 12-hydroxy stearic acid shall be packed in fiber or metal drums in accordance with Specification PPP-D-705, PPP-D-711, PPP-D-723 or PPP-D-729. Closure of the drums shall be made in accordance with the applicable specification. The gross weight shall not exceed 300 pounds.

5.2.2 Level C. The glyceryl ester of 12-hydroxy stearic acid shall be packed in accordance with 5.2.1 or as specified in contract or order.

5.3 Marking. Containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This material, glyceryl ester of 12-hydroxy stearic acid is intended for use in ammunition.

6.2 Ordering date. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type specified in the contract.
c. Provisions for submission of first article samples.
d. Acceptance and description sheets - Acceptance and description sheets shall be prepared for each lot in accordance with MIL-STD-1171.

6.3 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.4 The reactivity test will be performed by Picatinny Arsenal. Samples for this test shall be submitted in accordance with instructions of the Contracting Officer to Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-P, Dover, New Jersey 07801.

6.5 Supersession Data. This specification includes the requirements of Purchase Description PA-PD-721, dated 21 December 1955.

Custodian:
ARMY-PA

Preparing Activity:
ARMY-PA

Project Number: 6810-AA18

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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